# Framework for Organizing Viewsheds From Multiple Observation Points: Enabling Bidirectional Landscape Visibility Information Queries

Runzi Chen, Renjie Li, Pengcheng Deng, and Baolei Sun

***Abstract*—Viewshed analysis holds significant value in landscape assessment and management. However, current research faces dual limitations: inefficient organization of viewsheds from multiple observation points and inability to analyze bidirectional visibility relationships between multiple observation points and ground cells. This study develops a 3-D tensor (observation point–row–column) that integrates detailed visibility relationships between landscape units and ground cells and proposes a framework for organizing viewsheds from multiple observation points that supports bidirectional queries. The framework implements a three-stage optimization strategy: 1) decimal encoding of observation point dimension data, 2) a mapping mechanism from ground cells to observation point sets, and 3) run-length encoding-based organization of mapping data. Experimental results demonstrate that when using ASCII storage format, the framework achieves a 98% storage space reduction compared to conventional viewshed matrix methods while maintaining comparable query efficiency, with storage requirements similar to those of NetCDF format. Moreover, with 7-Zip compression, the framework achieves further space savings: a 40% reduction relative to viewshed matrix methods and a 70% reduction compared to NetCDF format. This research not only enriches the theory of multiple-observation-point viewshed analysis but also provides technical reference for geographic information product development, offering important application value for landscape conservation and planning.**

***Index Terms*—Data organization framework, geographic information system, landscape visibility analysis, viewshed analysis.**

Received 22 July 2025; revised 14 August 2025; accepted 21 August 2025. Date of publication 25 August 2025; date of current version 15 September 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 41471127 and in part by the Graduate Student Innovation Funding Project of Hebei Normal University under Grant XCXZZBS202409. *(Corresponding author: Renjie Li.)*

Runzi Chen and Pengcheng Deng are with the School of Geographical Science, Hebei Normal University, Shijiazhuang 050024, China (e-mail: 498363439@qq.com; d18060180655@163.com).

Renjie Li is with the School of Geographical Science, Hebei Normal University, Shijiazhuang 050024, China, also with the Hebei Technology Innovation Center for Remote Sensing Identification of Environmental Change, Shijiazhuang 050024, China, also with the GeoComputation and Planning Center of Hebei Normal University, Shijiazhuang 050024, China, and also with the Hebei Provincial Key Laboratory of Philosophy and Social Sciences "Research Laboratory of Geographic Big Data Computing and Resource Planning", Shijiazhuang 050024, China (e-mail: lrjgis@hebtu.edu.cn).

Baolei Sun is with the Tianjin Municipal Bureau of Planning and Natural Resources, Tianjin 300450, China (e-mail: stayreal9523@163.com).

Digital Object Identifier 10.1109/JSTARS.2025.3602482

## I. Introduction

LANDSCAPES are geographical spatial entities shaped through the interaction of natural and/or human factors, whose characteristics can be perceived and interpreted by humans [1]. In the process of landscape perception, vision, as humans' primary means of perception, can provide considerably more information than other senses, thus becoming the core pathway for interpreting and understanding landscapes [2], [3]. The term "viewshed" was introduced by Tandy in 1967, who drew on the analogy of a watershed and defined viewshed as the visible area from one or more observation points [4], [5]. With the development of geographic information system technology, viewshed analysis and visibility computation have become important tools for analyzing landscape value and function [6], [7].

Traditional viewshed analysis has developed along two main research directions. The first direction involves single-observation-point viewshed computation, where algorithms have matured significantly [8]. General intervisibility computation proceeds through three analytical steps.

1) Line-of-sight vector construction between observation and target points.
2) Digital terrain analysis for occlusion detection.
3) Binary visibility determination [9].

This workflow represents the paradigmatic approach in viewshed analysis. Several common algorithms for calculating the viewshed were first proposed, such as R3, R2, and XDraw. Regarding its brute force strategies, R3 is the slowest and most accurate, while R2 is faster than R3 as it makes estimations, and XDraw is the fastest among them but offers less accurate estimations [10]. These algorithms record viewsheds through Boolean raster format, with 1 and 0 representing visible and invisible areas, respectively [11]. In recent years, there have been significant advances in the research of viewshed calculation methods, for example, the PDERL [12], the HiPDERL [13], the XPDERL [14], and the view-tree method [15], which are constructed by establishing a proximity-direction-elevation coordinate system. These methods excel in computational accuracy and efficiency. However, when applied to landscape viewshed analysis, single-observation-point viewshed calculation requires abstracting the landscape into a single observation point, which has limitations in comprehensively capturing the landscape's visual characteristics and potential value. The second

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

direction encompasses multiple-observation-point viewshed analysis. Typical forms of multiple-point viewshed analysis include cumulative viewshed and total viewshed. The former creates a cumulative viewshed by overlaying individual viewsheds, where cell values indicate how many observation points can view that location [16]. This method has been applied in various fields, including visibility grading of 3-D architectural scenes [17], analysis of ancient town features [18], and optimization of viewing location selection [19]. The latter treats each cell in turn as a target. Every other cell in the region is considered in turn as an observer, and an intervisibility analysis is carried out to determine whether the target is visible [20]. This method is particularly useful for site selection decisions [21], [22], path planning [23], [24], urban planning [25], [26], and monitoring forest fires [27]. However, while cumulative viewsheds provide aggregate visibility information, they cannot precisely describe detailed visibility relationships between multiple observation points and ground cells.

A significant research gap remains: the lack of a method that can efficiently organize multiple-observation-point viewsheds [28] and support precise viewshed analysis from dual perspectives—"viewing" and "being viewed." This gap becomes particularly critical when analyzing complex landscapes with multiple observation points, given that understanding detailed visibility relationships is essential for comprehensive landscape assessment and management. For instance, tourists and landscape management authorities face two distinct analytical needs: 1) from a location perspective: identifying specific observation points visible from a given ground location (GL) to determine accessible landscape views; 2) from a landscape perspective: analyzing viewsheds of multiple designated observation points within the landscape system.

Unlike existing viewshed analysis methods, our goal is to efficiently organize numerous known individual observation point's viewsheds and lay the foundation for bidirectional visibility queries. This study proposes a multiobserver viewshed organization framework that achieves dual functionalities: 1) accurately characterizing visibility relationships between observation points and ground cells while 2) providing users with flexible configuration options through a series of optimization strategies.

## II. Methodology

Viewshed analysis based on regular-grid digital elevation models (RGDEMs) is the most straightforward and widely used approach [14]. Inspired by advances in spatiotemporal big data management in Earth sciences, particularly research on multidimensional spatiotemporal data organization structures [29], [30] and spatiotemporal cubes [31], [32], [33], [34], this study proposes a 3-D visibility model. Typically, digital elevation models (DEMs) can be identified by planar 2-D coordinates ($x$, $y$), where a pair of ($x$, $y$) determines a ground cell, and the cell value represents elevation. On this basis, we take landscape observation points as the first dimension and the coordinates ($x$, $y$) of the ground cell as the second and third dimensions, respectively, proposing a 3-D model of visibility relationships between landscape observation points and 2-D ground cells, denoted as visibility $= f$(point, $x$, $y$). In this model, visibility takes a value of 1 to indicate the existence of a visibility relationship and 0 to indicate the absence of a visibility relationship. This model can comprehensively describe accurate visibility relationships between multiple observation points of landscapes and ground cells.

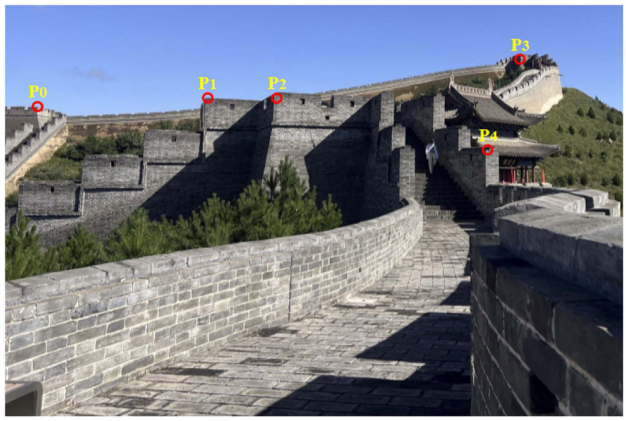



Fig. 1. SOP of the section of the great wall.

Before delving into the construction of a framework for multiple-observation-point viewsheds, we first introduce several fundamental concepts to ensure clarity in our subsequent discussion.

### A. Basic Concepts and Notation

1) *Set of Observation Points (SOPs)*: A set of points characterizing landscape features. The power set of the SOP is denoted as PSSOP.
2) *Set of Visible Observation Points (SVOPs)*: SOPs that are visible, which is a subset of the SOP.
3) *Code of Set of Visible Observation Points (CSVOPs)*: Through the establishment of an order for elements in the SOP, observation points appearing in the SVOP are represented by 1, and those not appearing are represented by 0, forming a bit string in accordance with this order.

*Example 1:* A section of the Great Wall can be described by five different observation points, p0–p4 (see Fig. 1), with an SOP of {p0, p1, p2, p3, p4}. The PSSOP is {{}, {p0}, {p1},. . ., {p0, p1, p2, p3, p4}}, where each set could potentially be an SVOP. If the order of observation points is defined as p0, p1, p2, p3, p4, then the CSVOP of {p0, p3, p4} is 10011.

*4) RGDEM*: Denoted as RGDEM $= \cup_{r=0}^{\text{row}-1} \cup_{c=\text{start}[r]}^{\text{end}[r]} \text{cell}_{rc}$, which corresponds to valid values in the ground DEM, where each cell has its own row and column indices; for example, $cell_{rc}$ represents the cell in row r and column c.

*5) GL*: Composed of one or more cells and represents a spatial subset of the RGDEM.

*6) Cell's SVOP ($SVOP_{cell}$):* The SVOP derived from a ground cell. GL's SVOP ($\text{SVOP}_{\text{GL}}$): The union of SVOPs from all cells within the GL.

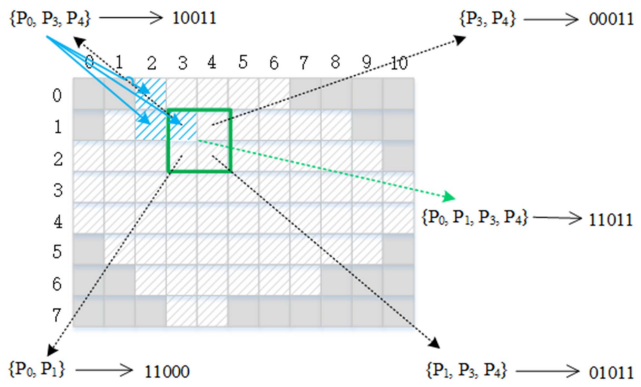



Fig. 2. Instance of RGDEM, $\mathrm{SVOP_{cell}}$, $\mathrm{SVOP_{GL}}$, and $\mathrm{GL_{SVOP}}$ (The set indicated by the black arrow is $\mathrm{SVOP_{cell}}$, the set indicated by the green arrow is $\mathrm{SVOP_{GL}}$, and the GL indicated by the blue arrow is $\mathrm{GL_{SVOP}}$.).

*7) SVOP's GL ( $GL_{SVOP}$ )*: The set of ground cells visible to the SVOP.

*Example 2:* As shown in Fig. 2, this case uses an 8-row by 11-column RGDEM, where each row is defined by a start column number and an end column number (e.g., row 6 spans from column 2 to column 7). Ground cells with a gray background represent missing data. Assuming the SOP matches Example 1, the SVOP for cell (1, 3) is $\{p_0, p_3, p_4\}$, with a corresponding CSVOP of 10011. The green box denotes a GL composed of four cells: (1, 3), (1, 4), (2, 3), and (2, 4). The GL's SVOP is the union of its constituent cells' SVOPs: $\{p_0, p_3, p_4\} \cup\!\!\!/ \{p_3, p_4\} \cup\!\!\!/ \{p_0, p_1\} \cup\!\!\!/ \{p_1, p_3, p_4\} = \{p_0, p_1, p_3, p_4\}$, represented by the CSVOP 11011. The $\mathrm{GL_{SVOP}}$of $\{v0, v3, v4\}$ is composed of cells (0, 2), (1, 2), and (1, 3). The decimal representation of the CSVOP(10011) is 25 ($2^0$+$2^3$+$2^4$).

### B. Framework Construction

The organization framework for multiple-observation-point viewsheds, as shown in Fig. 3, takes multiple single-observation-point viewsheds as input, integrates viewsheds from all observation points, and adopts four steps for optimized organization.

The "First" step generates a 3-D array based on the overlay of viewsheds from individual observation points. Similar to those in existing studies [35], [36], in this 3-D array, the first dimension ($z$-axis) represents observation points, while the second and third dimensions ($x$-, $y$-axes) represent ground cell coordinates. This method serves as a reference benchmark. Given 36 observation points, 36 corresponding viewsheds exist, with visibility represented as binary values (0 for invisible, 1 for visible). At viewshed coordinates (0,0), the visibility status of the 36 observation points could be 0,1,0,1,0,0,0,1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0, 0,1,1,1. The length of the 3-D array in the $z$-axis direction would be 36.

The "Second" step uses decimal encoding to encode $z$-axis values, creating a smaller 3-D array. In the 36-bit data in the above example, following the binary bit order from low to high, the first 32 bits are converted into an unsigned integer 394 (0,1,0,1,0,0,0,1,1,0,0,0,0,0,0,0,0,0,0,0, 0,0,0,0,0,0,0,0,0,0, 0,0,0,0,0,0 –> $2^1 + 2^3 + 2^7 + 2^8 = 394$) and the remaining 4 bits become an unsigned integer 14 (0,1,1,1 –> $2^1 + 2^2 + 2^3 = 14$). Through this conversion, the $z$-axis length of the 3-D array is reduced from 36 to 2, achieving the first optimization in storage.

The "Third" step constructs a mapping from RGDEM to PSSOP, as shown in Fig. 4, achieving the second optimization in storage. In this mapping, each cell corresponds to an SVOP (i.e., a subset of the SOP), and a set can be mapped to zero, one, or multiple cells. For sets with no corresponding cells, no cells have visibility relationships with that set, and such set information will not be stored. This mapping has the following two properties.

1) $\forall \quad \mathrm{GL_1} \subseteq \mathrm{GL_2} \quad \subseteq \mathrm{RGDEM}, \quad \mathrm{SVOP_{GL_1}} \quad \subseteq \mathrm{SVOP_{GL_2}}$.
2) $\forall\ \mathrm{sop_1} \subseteq \mathrm{sop_2} \in \mathrm{PSSOP}, \mathrm{GL_{SOP_2}} \subseteq \mathrm{GL_{SOP_1}}$.

In the implementation, a 2-D array of the same size as RGDEM is designed. The $z$-axis values (i.e., the decimal CSVOP) of each cell identified by the $x$, $y$ coordinates in the decimal-encoded 3-D array are traversed, deduplicated, and organized into a linear list with assigned indices, thereby achieving unique storage of decimal CSVOP. This approach is equivalent to uniquely linearizing the storage of the $z$-axis data. Finally, these indices are stored in the corresponding positions of the 2-D array, achieving the second storage optimization.

For example, this study supposes that 36 cells exist, each containing 2 unsigned integers on the $z$-axis, but among these 36 pairs, only 5 are distinct: (394, 14), (2345, 11), (234567, 4), (125, 2), (134567, 15). These five pairs are organized linearly and assigned indices from 0 to 4. Each element in the 2-D array then only needs to store one of these indices. As shown in the section between "Third" and "Fourth" in Fig. 3, the cells at coordinates (0,0), (0,1), (0,2), (1,0), and (1,1) have a decimal CSVOP value of (394, 14).

Run-length encoding (RLE) is a lossless data compression algorithm particularly effective for data containing long sequences of repeated values [37]. The algorithm operates by replacing consecutive identical values with (value: count) pairs. As demonstrated in the fourth processing step of Fig. 3, the 2-D array from row 0 to 5 is compressed as follows:

Row 0: 0:3, 3:3; Row 1: 0:2, 1:1, 2:2, 3:1; Row 2: 1:3, 2:2, 3:1; Row 3: 1:1, 2:3, 3:2; Row 4: 2:2, 4:2, 3:2; Row 5: 4:4, 3:2.

The "Fourth" step applies RLE to further organize the above 2-D array before persistent storage, achieving the third optimization in storage.

Based on this framework, this study performs bidirectional queries: 1) Given a GL, determine what landscape can be observed from that location, specifically identifying, which observation points characterizing the landscape are visible. 2) Given a subset of the SOPs that represents a particular landscape state, identify the GLs from which this state can be observed.

### C. Key Algorithms

Among the four steps of the framework, the "Third" step, which involves the second optimization, is the most complex. This subsection first introduces this process, then presents two

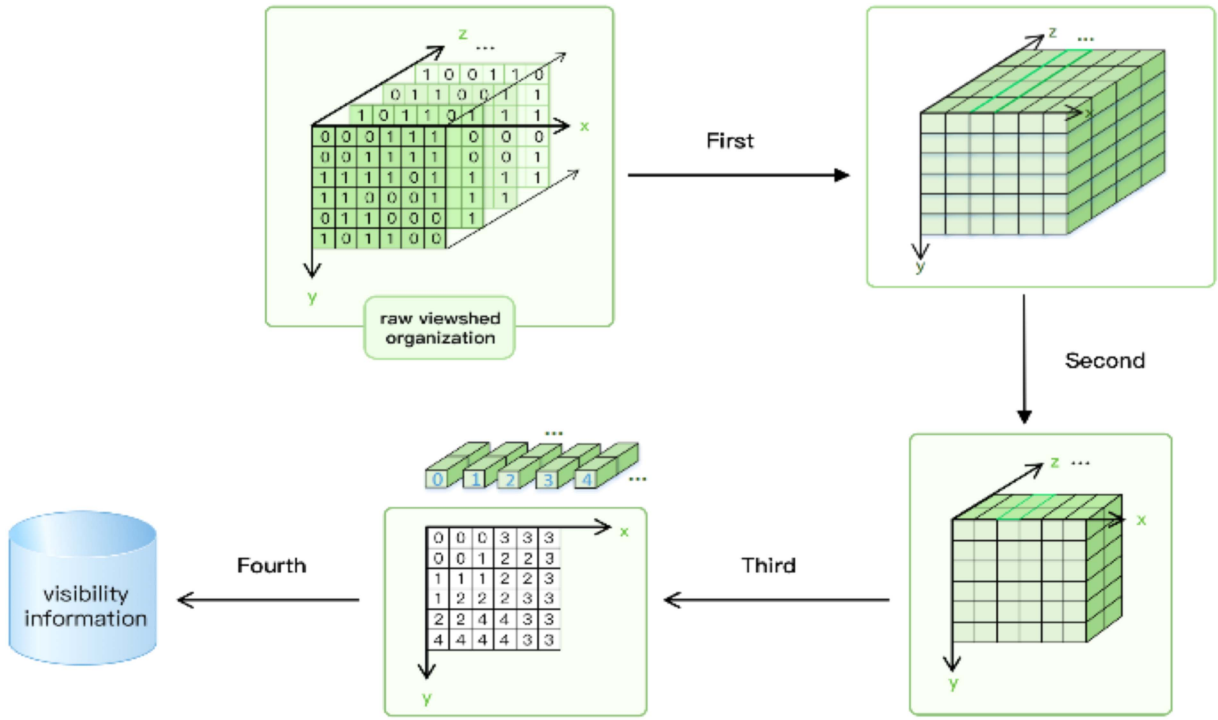



Fig. 3. Organization framework for multiple-observation-point viewsheds ("First": Overlay viewsheds from all observation points to form a 3-D array. "Second": Encode the observation point dimension in decimal representation. "Third": Construct a mapping from cells to decimal-encoded SVOP. "Fourth": Apply run-length encoding to the mapped information in the RGDEM.).

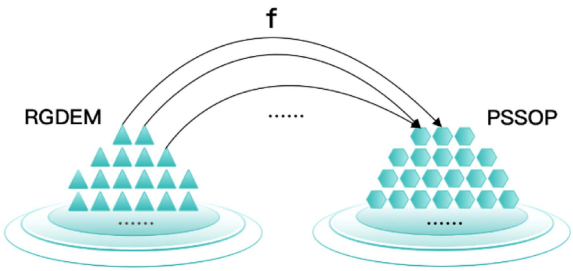



Fig. 4. Mapping from RGDEM to PSSOP (A triangle denotes a ground cell, and a hexagon represents an SVOP.).

TABLE I
EXPLANATIONS OF PARAMETERS IN ALGORITHMS

| No. | Parameter | Data type | Annotation |
|---|---|---|---|
| 1 | OptiData3 | Array | 3-D array after z-axis optimization |
| 2 | SVOP | Set | Set of visible observation points |
| 3 | DCSVOPL | List | List formed by the decimal CSVOP |
| 4 | Data2 | Array | Same size as that of RGDEM, mapping from RGDEM to PSSOP |
| 5 | GL | Set | Subset of RGDEM |
| 6 | $DCSVOP_{GL}$ | Array | Decimal CSVOP for GL |
| 7 | $GL_{SVOP}$ | Set | Set of ground cells for SVOP |

specific applications based on this framework: querying the SVOP for a GL and querying the GL for an SVOP. The parameters in these algorithms and their explanations are shown in Table I.

Algorithm 1 is used to implement the second optimization of the framework, which is the implementation of "*f*" in Fig. 4. Its input is the 3-D array OptiData3 after the first optimization, and it has two outputs: a 1-D list DCSVOPL (each element in the list is a decimal code of the SVOP) and a 2-D array Data2 (array element values are numbers of certain elements in the above list, reflecting the visibility relationship between ground cells and SVOPs). It consists of two main parts. In lines 2–8, a dictionary DCSVOPD is constructed, with values in the z-axis direction of the 3-D array as keys and tuples of row and column indices as values. This dictionary reflects the mapping from PSSOP to RGDEM, embodying the landscape perspective. In lines 9–14, the keys in the above dictionary are organized into a list, and the number of each key in the list is stored in the 2-D array corresponding to the key value's row–column tuples (multiple tuples may exist). This part implements the mapping from RGDEM to PSSOP, embodying the GL perspective.

Algorithm 2 is used to query the SVOP for GL. It takes three inputs: a GL (composed of coordinate tuples), the 2-D array Data2, and the list DCSVOPL output from Algorithm 1. The output is the SVOP for GL. The implementation process is as follows: Elements (row–column index tuples) are sequentially taken from GL. These elements are used to locate the index value of visible observation points in Data2. The index value is

**Algorithm 1:** Second Optimization.

Input: OptiData3
Output: DCSVOPL, Data2
1. DCSVOPL, DCSVOPD, Data2 ← Ø, Ø, Ø
2. for $i$ in len(OptiData3[0]): // Iterating through row indices of the location cell matrix
3. for $j$ in len(OptiData3[0][0]): // Iterating through column indices
4. DCSVOP ←OptiData3[:][$i$][$j$]
5. if DCSVOP in DCSVOPD:
6. add $(i, j)$ to the value of DCSVOP in DCSVOPD
7. else: // DCSVOP is not in the dictionary DCSVOPD
8. DCSVOPD[DCSVOP] ← $(i, j)$
9. $k \leftarrow 0$
10. for key in DCSVOPD.keys():
11. DCSVOPL[$k$] ← key
12. for $(i, j)$ in the value of DCSVOPD[key]:
13. Data2[$i$][$j$] ← $k$
14. $k \leftarrow k + 1$

utilized to locate SVOP in the list. Finally, $\text{SVOP}_{\text{GL}}$ is computed in accordance with concept (6).

**Algorithm 2:** Inquiring the Visibility of a Landscape via GL Perspective.

Input: GL, Data2, DCSVOPL
Output: $\text{SVOP}_{\text{GL}}$
1. $\text{DCSVOP}_{\text{GL}} \leftarrow \emptyset$
2. for $(i, j)$ in GL:
3. $k \leftarrow$ Data2[$i$][$j$]
4. $\text{DCSVOP}_{\text{GL}} \leftarrow \text{DCSVOP}_{\text{GL}} \cup\!\!\!\!\!\!\geqslant \text{DCSVOPL}[k]$ //concept 6
5. $\text{SVOP}_{\text{GL}} \leftarrow \text{DCSVOP}_{\text{GL}}$ //binary encoding and concept 3

Algorithm 3 is used to query the GL information for an SVOP. It takes three inputs: the SVOP, the 2-D array Data2, and the list DCSVOPL output from Algorithm 1. The output is $\text{GL}_{\text{SVOP}}$, the set of cells that have visibility relationships with SVOP.

**Algorithm 3:** Inquiring a GL via Landscape Perspective.

Input: SVOP, DCSVOPL, Data2
Output: $\text{GL}_{\text{SVOP}}$
1. Apply concept (3) with decimal encoding to transform SVOP into DCSVOP
2. $\text{GL}_{\text{SVOP}} \leftarrow \emptyset$
3. $k \leftarrow$ Obtaining the index of DCSVOP in DCSVOPL
4. for $i$ in len(Data2):
5. for $j$ in len(Data2[0]):
6. if $k ==$ Data2[$i$][$j$]:
7. $\text{GL}_{\text{SVOP}} \leftarrow \text{GL}_{\text{SVOP}} \cup\!\!\!\!\!\!\geqslant \{(i, j)\}$ // concept 7

For notational convenience, $M$ and $N$ denote the numbers of rows and columns of RGDEM, respectively.

Algorithm 1 consists of two parts, with the first part having a time complexity of $O(M \times N)$. The second part also has a time complexity of $O(M \times N)$, as it divides $M \times N$ cells into $k$ parts, where $k$ is the number of SVOPs, usually much smaller than $M \times N$. Therefore, the overall time complexity of Algorithm 1 is $O(M \times N)$. The space complexity of Algorithm 1 is $O(\max(M \times N, k \times \lceil \text{Num}/32 \rceil))$, where Num is the number of observation points of landscape, and $k$ is len(DCSVOPD.keys()), denoting the number of SVOPs.

For Algorithm 2, the time complexity is $O(\text{len(GL)})$, where len(GL) is the number of cells in GL. The space complexity is $O(\text{len(SOP)})$, where len(SOP) is the number of observation points.

For Algorithm 3, the time complexity is $O(M \times N)$. The space complexity is $O(\text{len}(\text{GL}_{\text{SVOP}}))$, where $\text{len}(\text{GL}_{\text{SVOP}})$ is the number of cells in $\text{GL}_{\text{SVOP}}$.

## III. Experiments and Results

### A. Case Study Region

China's famous Great Wall was built over a period of more than 2000 years, from the Warring States period to the Qing Dynasty [38]. Its total length exceeds 21 000 km. The experimental area in this study is located in Chengde City, Hebei Province, adjacent to Beijing, including the Jinshanling Great Wall scenic area. The area uses ALOS 12.5 m-resolution DEM as basic terrain data, with a grid size of 1775 × 2059 and elevation values ranging from 143 to 1309 m. The Great Wall landscape is 10 000 m long, composed of 1542 observation points.

### B. Data Preprocessing

We first randomly select 100 points from the 1542 points along the Great Wall as landscape observation points. Then, we use a Python script to calculate the viewshed for each observation point. Finally, GDAL [39] is employed to read each viewshed to form a 2-D array of 0 s and 1 s. The processed visibility data consist of 100 2-D arrays, each containing 1775 rows and 2059 columns.

### C. Experimental Design

This section verifies the scalability and usability of the proposed framework. We first extract subarrays containing 1, 1.5, 2, 2.5, and 3 million elements from the 100 1,775 × 2,059 2D arrays and then conduct the following four experiments.

1) Common multidimensional raster data file formats include NetCDF, GRIB, HDF, and CRF, each with distinct characteristics in structure, functionality, and applicability [40]. Network common data form (NetCDF) is particularly notable as a self-describing, machine-independent, array-based scientific data format [41], [42] that offers excellent portability, cross-platform compatibility, and open-source environment support [43]. The storage scalability of different organizational architectures was validated for multiple-observation-point viewsheds (corresponding to the "First" through "Fourth" methods in Fig. 3). All experimental datasets were stored on local disks in plain

TABLE II
SEMANTIC CLASSIFICATION OF GREAT WALL LANDSCAPE OBSERVATION POINTS

| Functional semantics | Numbers of points | ID |
|---|---|---|
| Beacon Tower | 2 | 88, 89 |
| Enemy Station | 7 | 6, 23, 87, 90–92, 97 |
| Wall | 91 | 0–5, 7–22, 24–86, 93–96, 98–99 |

TABLE III
STORAGE SPACE REQUIREMENTS OF FIVE ORGANIZATIONAL APPROACHES (SETTING THE GL TO CONSIST OF 1 MILLION CELLS. DATA ARE MEASURED IN MB)

| Number of points | Benchmark | Second | Third | Fourth | NetCDF |
|---|---|---|---|---|---|
| 10 | 19.721 | 2.003 | 3.94 | **0.059** | 0.125 |
| 40 | 78.882 | 5.436 | 5.102 | **0.958** | 1 |
| 70 | 138.043 | 8.185 | 6.755 | **2.104** | 1.639 |
| 100 | 197.204 | 10.901 | 7.495 | **3.028** | 2.031 |

The bold values highlight the final results of our proposed framework.

text format to verify the framework's core organizational functionality. For comparative analysis, a parallel storage solution utilizing the NetCDF format was concurrently implemented.

a) The set of GLs is fixed to contain 1 million cells (rows × columns = 1 million). The storage space requirement scalability of visibility information between Great Wall landscapes represented by 10, 40, 70, and 100 observation points and the GL is verified.
b) The number of observation points is fixed at 70. The storage space requirement scalability of visibility information between Great Wall landscapes represented by 70 observation points and GLs containing 1, 1.5, 2, 2.5, and 3 million cells is verified.

2) The time costs of three optimization processes ("Second," "Third," and "Fourth" in Fig. 3) are verified. Similar to those in the first experiment, the number of cells in the GL and the number of observation points are fixed. The time overhead for each optimization process is calculated.
3) The time costs of bidirectional visibility information queries based on different organization methods of multiple-observation-point viewsheds are verified.
   a) The number of observation points is fixed at 70. The time costs of querying visible observation points from GLs composed of 10 000, 50 000, 100 000, 150 000, and 200 000 cells are verified.
   b) The GL is fixed to contain 1 million grids. The time costs of querying the GL for sets of visible observation points containing 10, 20, 30, 40, and 50 observation points are verified.
4) The framework's usability is verified. On the basis of visibility data between the GL containing 1 million cells and 100 observation points representing the Great Wall, two types of queries are conducted.
   a) *Query of the GL of SVOP*: The 100 observation points are classified in accordance with the Great Wall's semantic functions (see Table II). The GL that can only see 2 beacon towers and the GL that can see a straight-line section of the Great Wall formed by 18 observation points are separately queried.
   b) *Query of the SVOP of GL*: GL A within the Jinshanling Great Wall scenic area and GL B along the road outside the scenic area are selected. The observation points (out of 100) that can be seen from these 2 GLs are determined.

TABLE IV
STORAGE SPACE REQUIREMENTS OF FIVE ORGANIZATIONAL APPROACHES (SETTING THE SOP TO CONTAIN 70 OBSERVATION POINTS. DATA ARE MEASURED IN MB)

| Size of GL | Benchmark | Second | Third | Fourth | NetCDF |
|---|---|---|---|---|---|
| 1 | 138.043 | 8.185 | 6.755 | **2.104** | 1.639 |
| 1.5 | 217.652 | 13.056 | 10.549 | **3.101** | 2.577 |
| 2 | 273.999 | 16.721 | 13.267 | **3.837** | 3.78 |
| 2.5 | 342.073 | 20.835 | 16.471 | **4.534** | 4.523 |
| 3 | 410.813 | 25.076 | 19.674 | **5.216** | 5.301 |

The bold values highlight the final results of our proposed framework.

These experiments aim to comprehensively evaluate the performance and practicality of the proposed framework under different data scales and query scenarios. They test storage efficiency, optimization cost, query speed, and usability in practical application scenarios.

### D. Experimental Environment

This study employs a dual-language implementation architecture: the core framework is developed in C/C++, with a complementary Python-based NetCDF storage module implemented for comparative evaluation. All experiments have been conducted on a workstation equipped with an Intel(R) Core (TM) i7-7700HQ quad-core 2.8 GHz CPU, 16 GB RAM, and the Windows 10 × 64 Home operating system.

### E. Results and Analysis

Among the four organizational approaches in the framework, the first method adopts existing spatiotemporal data organization techniques, which we employ as the benchmark.

1) Storage space required for different organization methods of multiple-observation-point viewsheds.
   a) The GL is fixed to contain 1 million grids. The storage space for visibility information between Great Wall landscapes represented by 10, 40, 70, and 100 observation points and the GL is compared (see Table III).
   b) The number of observation points is fixed at 70. The storage space for visibility information between Great Wall landscapes represented by 70 observation points and GLs containing 1, 1.5, 2, 2.5, and 3 million cells is compared (see Table IV).

The storage space size required by each organizational approach in the framework shows a linear relationship with the number of observation points and the number of cells in GL (see Tables III and IV). Each optimization can further save space. In the first optimization ("Second"), converting data values 0 and 1 into integer encoding saves over 90% space. The second

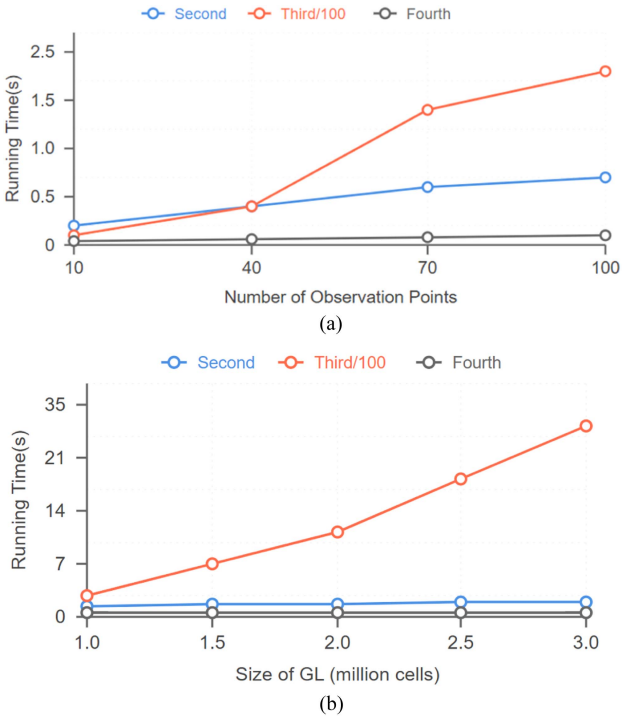



Fig. 5. Running time for three optimization processes (The running time of the "Third" optimization is divided by 100 for the yellow line). (a) Setting the GL to consist of 1 million cells. (b) Setting the SOP to contain 70 observation points.

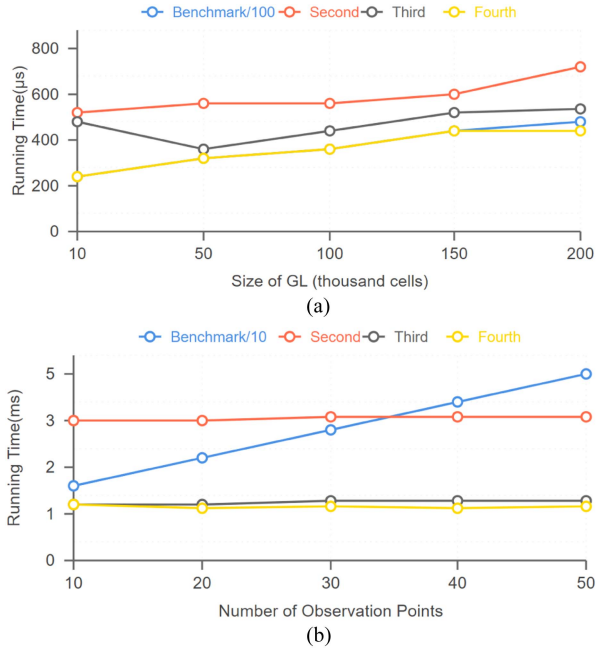



Fig. 6. Running time for two queries (The blue line in a is the result of the time for querying on the benchmark divided by 100; in b, it is divided by 10). (a) Time cost of querying landscape visibility. (b) Time cost of querying GL.

optimization ("Third") requires a certain number of observation points to be effective because the 2-D array representing visibility relationships needs to store the numbers of visible observation point sets. We set these numbers as integer type; hence, when fewer observation points exist, the advantage of this compression is not obvious. The third optimization ("Fourth") further optimizes cells with the same SVOP. These three optimizations together can save 98% of storage space.

Although ASCII encoding is generally inefficient for storing geospatial data, our proposed method achieves storage efficiency comparable to that of the NetCDF format due to the compression techniques employed.

2) Time overhead of three optimizations.
   a) The GL is fixed to contain 1 million grids. The time costs of the "Second," "Third," and "Fourth" optimizations in processing visibility information between landscapes containing 10, 40, 70, and 100 observation points and the GL are provided [see Fig. 5(a)].
   b) The number of observation points is fixed at 70. The time costs of the "Second," "Third," and "Fourth" optimizations in processing visibility information between observation points and the GL containing 1, 1.5, 2, 2.5, and 3 million grids are provided [see Fig. 5(b)].

Note: For ease of comparison, the time cost of the "Third" optimization is divided by 100 before comparison.

The time costs of the three optimizations show a linear relationship with the size of the GL and the number of landscape observation points (see Fig. 5). The "Third" optimization has a higher time cost because this study employs a search method for redundancy elimination.

3) Time cost of two perspective queries.

Owing to fast query speeds, each query is performed 1000 times to amplify the differences.

a) *Landscape perspective queries:* Under the four visibility information organization methods in the framework, the time costs of querying SVOP from GLs composed of 10 000, 50 000, 100 000, 150 000, and 200 000 cells are provided. For ease of comparison, the query time under the benchmark is divided by 100 [see Fig. 6(a)].
b) *GL perspective queries:* Under the four visibility information organization methods in the framework, the time costs of querying GL for SVOPs containing 10, 20, 30, 40, and 50 observation points are provided. For ease of comparison, the query time under the benchmark is divided by 10 [see Fig. 6(b)].

For the landscape perspective queries, the query time shows a linear relationship with the size of the query area [see Fig. 6(a)]. The first method (benchmark) takes more time for queries because each cell corresponds to 70 data points, and the 3-D array uses lower index priority storage. The query times after the "Third" and "Fourth" steps are roughly the same because these two optimizations only affect the data file size while keeping the memory structure unchanged, resulting in similar query times.

For the GL perspective queries, when visibility information organization is implemented using the benchmark, the query time only depends on the number of observation points. After optimization, query time becomes constant because optimization processes 32 observation points together. The query times after the "Third" and "Fourth" optimizations are shorter than

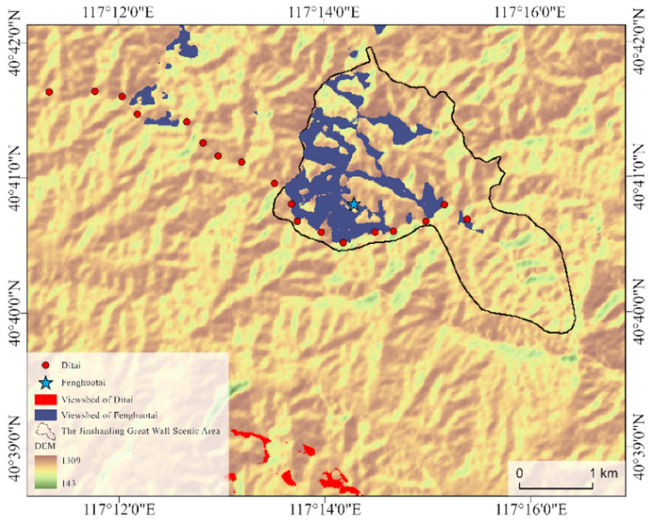



Fig. 7. Instances of querying GL from a landscape.

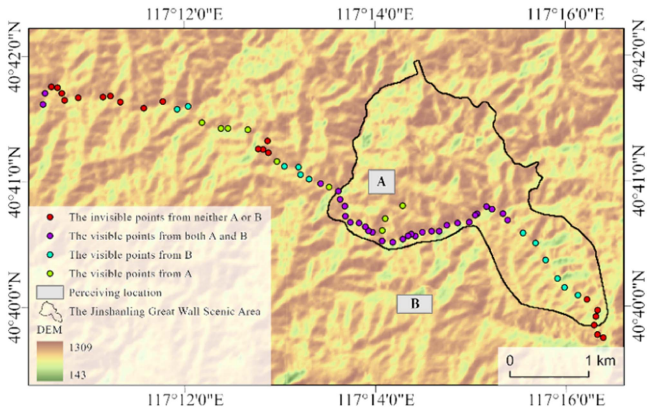



Fig. 8. Instances of querying landscape visibility from GL.

that after the "Second" optimization because the SVOPs have been deduplicated [see Fig. 6(b)].

4) Framework usability verification.

To visualize GL queries, we use red-filled pentagons to represent beacon towers, with red areas indicating the GL from which these two beacon towers are visible. Eighteen blue dots represent a straight section of the Great Wall, with blue areas indicating the GL from which the straight section is visible (see Fig. 7). This representation helps analyze the strategic value of the Great Wall, such as the coverage of beacon towers.

For visualizing landscape observation point queries, three different colors are used to represent the following situations (see Fig. 8): observation points visible only from GL A, observation points visible only from GL B, observation points visible from both GLs A and B. This visualization scheme helps compare visibility differences between different GLs (such as GLs A and B), providing a scientific basis for tourism planning and viewing point layout.

## IV. Discussion

Research on organizing visibility relationship data between landscapes with multiple observation points and GL remains relatively scarce. This study constructs and implements a multiple-observation-point viewshed organization framework focused on bidirectional querying of landscape visibility information, enabling users to conveniently analyze landscape visibility from GL and landscape perspectives. The following discussion unfolds from methodological and empirical dimensions.

### A. Methodological Analysis

Given that each observation point's viewshed is known, this study constructs a framework for organizing multiple-observation-point viewsheds. We design related data structures and algorithms to implement each component of the framework sequentially. Each component can be directly used to store detailed visibility information between multiple observation points and ground cells, and users can choose specific optimization steps on the basis of actual situations. Details are given below.

1) The "Second" step converts each cell's CSVOP (a sequence composed of 0 and 1 s) into integer encoding, shortening the $z$-axis of the 3-D array. This reduction is applied universally to all cells without additional conditions. The more observation points there are, the better this step performs. For visibility information between the GL at the million-cell level and landscapes with 100 observation points, running time is only at the second level. This condition provides a good reference for organizing multiple-observation-point viewsheds.
2) The main purpose of the "Third" step is to eliminate duplicate parts of SVOP for cells, i.e., storing identical data in the $z$-axis direction only once. It is suitable for scenarios in which the first law of geography is obvious in visibility information. The experiment used a search method, resulting in increased algorithmic time cost. If a hash method is adopted, the time cost would decrease, but it might require additional memory space. The main advantage of the "Third" step lies in improving bidirectional query efficiency. This optimization achieves higher query efficiency than 3-D array storage because of the storage characteristics of 3-D arrays prioritizing lower dimensions. Furthermore, although this optimization takes longer, i.e., at the minute level, this overhead is one time, meaning it only needs to be done once to optimize the data structure for multiple-observation-point viewsheds. These optimized data can be queried infinitely by users, making the cost effective. When excessive observation points exist or if the data resolution is extremely high, the repetition rate of SVOP might not be very high, making the advantages of this organization inconsiderably obvious.
3) The "Fourth" step is related to the "Third" step, compressing storage for GL cells with the same SVOP, further reducing visibility information data volume. This optimization is mainly used in data persistence, providing some reference for the production of large-scale visibility products.

TABLE V
COMPRESSED STORAGE SPACE COMPARISON OF FOUR ORGANIZATIONAL APPROACHES (USING WINDOWS 7-ZIP, SETTING THE GL TO CONSIST OF 1 MILLION CELLS. DATA ARE MEASURED IN KB.)

| Number of points | Benchmark | Second | Third | Fourth | NetCDF |
|---|---|---|---|---|---|
| 10 | 23 | 13 | 14 | **10** | 49 |
| 40 | 385 | 227 | 223 | **194** | 804 |
| 70 | 638 | 366 | 434 | **408** | 1280 |
| 100 | 811 | 464 | 572 | **552** | 1600 |

The bold values highlight the final results of our proposed framework.

TABLE VI
COMPRESSED STORAGE SPACE COMPARISON OF FOUR ORGANIZATIONAL APPROACHES (USING WINDOWS 7-ZIP, SETTING THE SOP TO CONTAIN 70 OBSERVATION POINTS. DATA ARE MEASURED IN KB)

| Size of GL | Benchmark | Second | Third | Fourth | NetCDF |
|---|---|---|---|---|---|
| 1 | 638 | 366 | 434 | **408** | 1218 |
| 1.5 | 1051 | 568 | 652 | **604** | 2091 |
| 2 | 1375 | 725 | 819 | **755** | 2775 |
| 2.5 | 1696 | 874 | 972 | **896** | 3300 |
| 3 | 2025 | 1026 | 1129 | **1037** | 3879 |

The bold values highlight the final results of our proposed framework.

The framework effectively solves computational efficiency and storage issues in large-scale landscape viewshed analysis through effective data structures and multiple-level optimization strategies. It not only optimizes data processing but also maintains data integrity, providing an effective and flexible tool for landscape analysis and related research.

For operational convenience, all experimental data were initially stored in ASCII format. Subsequently, NetCDF format was adopted. We then compressed each dataset using 7-Zip (see Tables V and VI). The three comparative methods ("Second," "Third," and "Fourth") showed comparable storage requirements, achieving a 40% reduction from the benchmark. Our method provided an additional 70% reduction beyond 7-Zip-compressed NetCDF, indicating that NetCDF's inherent data-type-level optimization leaves minimal room for conventional compression techniques. Note that compressed data require decompression before analysis.

### B. Empirical Research

The effective organization of visibility information between the Jinshanling Great Wall landscape and surrounding GL provides new insights for the protection and utilization of Great Wall cultural heritage in the context of Chinese cultural park development. In the experiments, we randomly selected 100 observation points representing the Great Wall, including 1 beacon tower, 18 enemy towers, and some wall bodies containing key information, while 2 representative locations in the study area were chosen. We calculated sets of visible observation points on the basis of locations with special significance and specified locations.

Although we only conducted empirical research on a part of the Great Wall, the framework constructed in this study can be applied to any viewshed analysis between landscapes and RGDEM. This study applies advanced data analysis techniques to cultural heritage research, providing data-driven decision support for landscape protection and planning, methodological reference for research on other types of cultural heritage or natural landscapes, and data-based decision tools for policymakers and planners.

### C. Framework Limitations

The storage space required for a single-observation-point viewshed is proportional to $M \times N$ terrain grid cells, where $M$ and $N$ represent the number of rows and columns, respectively. For $n$ observation points, the original storage space is proportional to $n \times M \times N$.

Our proposed framework effectively organizes multiple-observation-point viewsheds. However, it has two limitations: First, when n is particularly large, indicating fine-grained landscape representation, subtle differences in the landscape states visible from each grid cell may reduce the compression effectiveness of the subsequent two steps. Second, high terrain resolution with fine-grained grids may also diminish the compression performance of the latter two steps.

## V. CONCLUSION AND FUTURE WORK

This study, starting from the perspective of human–environment relationships and focusing on bidirectional querying of landscape visibility information, proposes and implements an organization framework for continuously optimizing visibility information between multiple observation points and ground cells.

The main contributions of this study are as follows.

1) A visibility model based on multiple observation points for landscapes and each cell is constructed. This model can comprehensively and clearly represent visibility relationships between all observation points and visual locations, providing model support for further deep value mining of visual landscape resources and offering a new perspective for landscape viewshed analysis research.
2) A framework for integrating and organizing viewsheds from multiple observation points is proposed and implemented. This framework can organize and continuously optimize visibility information between multiple observation points and ground cells. After implementing each component of the framework, this study objectively analyzes the costs and applicable scenarios of components. Landscape visual analysis users can choose appropriate components as their analysis tools on the basis of actual situations. The entire framework can provide useful reference for the production of related geographic data products in the geographic information industry.
3) Visibility information between the Jinshanling Great Wall landscape and its surrounding GL is organized, helping understand the differences in visual perception of the Jinshanling landscape from different GLs and providing data support for further development of the Jinshanling Great Wall landscape. From the perspective of visual perception,

this organization provides reference for decision-making by landscape management departments.

This research not only achieves efficient storage of landscape visibility information but also provides new perspectives for landscape visual analysis and management by transforming complex spatial visibility relationships into operable data structures. However, this study focuses on single-scale analysis, which limits its applicability in comprehensive landscape assessment. With advances in remote sensing technology and growing demands for high spatiotemporal resolution data [44], artificial intelligence techniques [45], [46] can be employed to improve DEM resolution [47], [48]. Future research should explore multiscale representation of landscapes [49], which will help achieve accurate representation of landscape visual resources and identify key observation points requiring special protection. This framework's interdisciplinary research value can be further enhanced by integrating concepts and methods from remote sensing, landscape ecology, and computer science.

## REFERENCES


[1] F. M. Wartmann, J. Frick, F. Kienast, and M. Hunziker, "Factors influencing visual landscape quality perceived by the public. Results from a national survey," *Landsc. Urban Plan.*, vol. 208, 2021, Art. no. 104024.

[2] M. Sharafatmandrad and A. Khosravi Mashizi, "Visual value of rangeland landscapes: A study based on structural equation modeling," *Ecol. Eng.*, vol. 146, 2020, Art. no. 105742.

[3] R. J. Henriques et al., "Soils and landscapes of Marajó island, Brazilian Amazonia: Holocene evolution, geoarchaeology and climatic vulnerability," *Environ. Earth Sci.*, vol. 81, no. 9, 2022, Art. no. 254.

[4] N. C. Inglis, J. Vukomanovic, J. Costanza, and K. K. Singh, "From viewsheds to viewscapes: Trends in landscape visibility and visual quality research," *Landsc. Urban Plan.*, vol. 224, 2022, Art. no. 104424.

[5] K. Ruzickova, J. Ruzicka, and J. Bitta, "A new geographic information system (GIS)-compatible methodology for visibility analysis in digital surface models of earth sites," *Geosci. Front.*, vol. 12, no. 4, 2021, Art. no. 101109.

[6] S. Hincks and R. Johnston, "The ephemerality of prominence: A geospatial analysis of acoustic affordances in a hillfort landscape," *J. Anthropol. Archaeol.*, vol. 67, 2022, Art. no. 101423.

[7] M. Haughton, "Connecting Posthumanist thinking with GIS practice: Explorations of a prehistoric heathland landscape in Jutland, Denmark," *J. Archaeol. Method Theory*, vol. 31, no. 1, pp. 227–250, 2024.

[8] Y. Wang and W. Dou, "A fast candidate viewpoints filtering algorithm for multiple viewshed site planning," *Int. J. Geogr. Inf. Sci.*, vol. 34, no. 3, pp. 448–463, 2020.

[9] G. Zhu, J. Li, J. Wu, M. Ma, L. Wang, and N. Jing, "HiXDraw: An improved XDraw algorithm free of chunk distortion," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 3, 2019, Art. no. 153.

[10] Y. Zhang, M. Ma, J. Li, A. Yang, Q. Jia, and Z. Liu, "Efficient and fine-grained viewshed analysis in a three-dimensional urban complex environment," *Environ. Model. Softw.*, vol. 188, 2025, Art. no. 106359.

[11] P. F. Fisher, "Algorithm and implementation uncertainty in viewshed analysis," *Int. J. Geogr. Inf. Sci.*, vol. 7, no. 4, pp. 331–347, 1993.

[12] C. Wu, L. Guan, Q. Xia, G. Chen, and B. Shen, "PDERL: An accurate and fast algorithm with a novel perspective on solving the old viewshed analysis problem," *Earth Sci. Informat.*, vol. 14, pp. 619–632, 2021.

[13] H. Cheng and W. Dou, "HiPDERL: An improved implementation of the PDERL viewshed algorithm and accuracy analysis," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 10, 2022, Art. no. 506.

[14] L. Guan, C. Wu, Q. Xia, G. Chen, and A. Li, "Fast approximate viewshed analysis based on the regular-grid digital elevation model: X-type partition proximity-direction-elevation spatial reference line algorithm," *Comput. Geosci.*, vol. 167, 2022, Art. no. 105213.

[15] Z. Wang, L. Xiong, Z. Guo, W. Zhang, and G. A. Tang, "A view-tree method to compute viewsheds from digital elevation models," *Int. J. Geogr. Inf. Sci.*, vol. 37, no. 1, pp. 68–87, 2023.

[16] S. D. Quinn, "What can we see from the road? Applications of a cumulative viewshed analysis on a US state highway network," *Geogr. Helv.*, vol. 77, no. 2, pp. 165–178, 2022.

[17] Y. Chen and J. Chen, "A parallel multipoint viewshed analysis method for urban 3D building scenes," *Trans. GIS*, vol. 25, no. 4, pp. 2010–2028, 2021.

[18] G. Openshaw and B. C. Chamberlain, "Optimizing viewpoint selection for route-based experiences: Assessing the role of viewpoints on viewshed accuracy," *Land*, vol. 11, 2022, Art. no. 1324.

[19] T. Zhang, M. Yan, X. Yu, and B. Liu, "Visual assessment of historic landmarks based on GIS and survey: A study of view and viewing of Tiger Hill in Suzhou, China," *J. Asian Archit. Build. Eng.*, vol. 23, no. 3, pp. 1079–1093, 2024.

[20] Y. H. Kim, S. Rana, and S. Wise, "Exploring multiple viewshed analysis using terrain features and optimisation techniques," *Comput. Geosci.*, vol. 30, no. 9, pp. 1019–1032, 2004.

[21] Y. Yao et al., "A site selection framework for urban power substation at micro-scale using spatial optimization strategy and geospatial big data," *Trans. GIS*, vol. 27, no. 6, pp. 1662–1679, 2023.

[22] G. Tang et al., "An efficient and precise multi-candidate viewpoint filtering algorithm for terrain viewshed selection," *Int. J. Geogr. Inf. Sci.*, vol. 39, no. 2, pp. 400–421, 2025.

[23] X. Qu, X. Xiao, X. Zhu, H. Wu, J. Gong, and D. Li, "6Ds-HLM: Hierarchical models of road traffic crash influencing factor analysis considering the urban built environment of intersection visibility," *Int. J. Appl. Earth Obs. Geoinf.*, vol. 123, 2023, Art. no. 103472.

[24] C. Zhang, Y. Wang, H. Cheng, and W. Dou, "Spatial semantic expression of terrain viewshed: A data mining method," *ISPRS Int. J. Geo-Inf.*, vol. 14, no. 3, 2025, Art. no. 113.

[25] C. Lehto, M. Hedblom, A. Filyushkina, and T. Ranius, "Seeing through their eyes: Revealing recreationists' landscape preferences through viewshed analysis and machine learning," *Landsc. Urban Plan.*, vol. 248, 2024, Art. no. 105097.

[26] Z. Tang et al., "A method for identifying urban road intersections considering the merging of parallel lanes," *Int. J. Geogr. Inf. Sci.*, vol. 39, no. 10, pp. 2333–2361, 2025.

[27] T. Yu, L. Xiong, M. Cao, Z. Wang, Y. Zhang, and G. A. Tang, "A new algorithm based on Region partitioning for filtering candidate viewpoints of a multiple viewshed," *Int. J. Geogr. Inf. Sci.*, vol. 30, no. 11, pp. 2171–2187, 2016.

[28] X. Yang, N. Fox, D. Van Berkel, and M. Lindquist, "Viewscape: An R package for the spatial analysis of landscape perception and configurations in viewsheds of landscapes," *SoftwareX*, vol. 26, 2024, Art. no. 101662.

[29] P. Baumann, "A general conceptual framework for multi-dimensional spatio-temporal data sets," *Environ. Model. Softw.*, vol. 143, 2021, Art. no. 105096.

[30] R. Béjar, J. Lacasta, F. J. Lopez-Pellicer, and J. Nogueras-Iso, "Discrete global grid systems with quadrangular cells as reference frameworks for the current generation of earth observation data cubes," *Environ. Model. Softw.*, vol. 162, 2023, Art. no. 105656.

[31] J. Phuong et al., "Automated retrieval, preprocessing, and visualization of gridded hydrometeorology data products for spatial-temporal exploratory analysis and intercomparison," *Environ. Model. Softw.*, vol. 116, pp. 119–130, 2019.

[32] F. Gao et al., "A multi-source spatio-temporal data cube for large-scale geospatial analysis," *Int. J. Geogr. Inf. Sci.*, vol. 36, no. 9, pp. 1853–1884, 2022.

[33] M. Iqbal, M. Lissandrini, and T. B. Pedersen, "A foundation for spatio-textual-temporal cube analytics," *Inf. Syst.*, vol. 108, 2022, Art. no. 102009.

[34] L. Dong et al., "A novel approach for cloud-free MODIS NDSI reconstruction on the Tibetan plateau combining spatiotemporal cube and environmental features," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4300814.

[35] M. Lu, M. Appel, and E. Pebesma, "Multidimensional arrays for analysing geoscientific data," *ISPRS Int. J. Geo-Inf.*, vol. 7, 2018, Art. no. 313.

[36] J. Zhou, J. Ben, Q. Liang, X. Huang, and J. Ding, "A general modeling scheme for spatiotemporal DGGS with emphasis on encoding and operating multiscale time grids," *Trans. GIS*, vol. 28, no. 5, pp. 1130–1155, 2024.

[37] K. T. Chang, *Introduction to Geographic Information Systems*, 9th ed. New York, NY, USA: McGraw-Hill, 2016.

[38] H. Xu, F. Chen, and W. Zhou, "A comparative case study of MTInSAR approaches for deformation monitoring of the cultural landscape of the Shanhaiguan section of the Great Wall," *Herit. Sci.*, vol. 9, no. 1, pp. 1–15, 2021.

[39] E. Rouault et al., 2025. GDAL (v3.10.1). Zenodo, doi: 10.5281/zenodo.14639689.
[40] G. Ravazzani, "MOSAICO, a library for raster based hydrological applications," *Comput. Geosci.*, vol. 51, pp. 1–6, 2013.
[41] T. Gan, D. G. Tarboton, J. S. Horsburgh, P. Dash, R. Idaszak, and H. Yi, "Collaborative sharing of multidimensional space-time data in a next generation hydrologic information system," *Environ. Modelling Softw.*, vol. 129, Jul. 2020, Art. no. 104706.
[42] S. Nativi, J. Caron, E. Davis, and B. Domenico, "Design and implementation of netCDF markup language (NcML) and its GML-based extension (NcML-GML)," *Comput. Geosci.*, vol. 31, no. 9, pp. 1104–1118, Nov. 2005.
[43] S. Kothe, R. Hollmann, U. Pfeifroth, C. Träger-Chatterjee, and J. Trentmann, "The CM SAF R toolbox—A tool for the easy usage of satellite-based climate data in NetCDF format," *ISPRS Int. J. Geo-Inf.*, vol. 8, no. 3, 2019, Art. no. 109.
[44] H. Shen, Z. Qiu, L. Yue, and L. Zhang, "Deep-learning-based super-resolution of video satellite imagery by the coupling of multiframe and single-frame models," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612114.
[45] A. Boutayeb, I. Lahsen-Cherif, and A. E. Khadimi, "When machine learning meets geospatial data: A comprehensive GeoAI review," *IEEE J. Sel. Top. Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13135–13191, 2025.
[46] Y. Duan et al., "LCIRE-net: Lightweight cross-modal information interaction for road feature extraction from remote sensing images and GPS trajectory/LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 2000418.
[47] C. Zhu, Y. Liu, S. Huang, and F. Wang, "Taming a diffusion model to revitalize remote sensing image super-resolution," *Remote Sens.*, vol. 17, 2025, Art. no. 1348.
[48] J. Sun, H. Shen, Q. Yuan, and L. Zhang, "Super-resolution for remote sensing imagery via the coupling of a variational model and Deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5611119.
[49] Z. He, Z. Wang, Z. Xie, L. Wu, and Z. Chen, "Multiscale analysis of the influence of street built environment on crime occurrence using street-view images," *Comput. Environ. Urban.*, vol. 97, 2022, Art. no. 101865.



**Runzi Chen** was born in Xinyang, Henan, China, in 1981. He received the B.S. degree in computer science and technology from Xinyang Normal University, Xinyang, China, in 2006, and the M.S. degree in software and theory of computer from Central China Normal University, Wuhan, China, in 2009. He is currently working toward the Ph.D. degree in cartography and geographic information systems with the School of Geography, Hebei Normal University, Shijiazhuang, China.

His current research interests include spatial data organization and mining, visibility analysis, and AI for geocomputation.



**Renjie Li** was born in Shijiazhuang, Hebei, China, in 1975. He received the Ph.D. degree in ecology from Hebei Normal University, Shijiazhuang, China, in 2013.

He is currently a Professor and Ph.D. Supervisor with the School of Geography, Hebei Normal University, and serves as a Director of the Hebei Geography Science Experimental Teaching Center. He is a Council Member of the Geographical Society of China and Associate Secretary-General of the Hebei Geographical Society. He has led and completed three National Natural Science Foundation projects, authored or coauthored more than 60 academic papers, obtained three software copyrights, and edited five GIS professional textbooks. His research interests include social media geographic big data mining, humanities and social applications of GIS and remote sensing, volunteered geographic information and tourism geography, with a primary focus on GIS and tourism geography research.



**Pengcheng Deng** was born in Sanming, Fujian, China, in 2000. He received the B.S. degree in urban and rural planning from Fujian University of Technology, Fuzhou, China, in 2023. He is currently working toward the M.S. degree in resources and environment with the School of Geography, Hebei Normal University, Shijiazhuang, China.

His research interests include computational studies of three-dimensional landscape visual perception.



**Baolei Sun** was born in Wuqing, Tianjin, China, in 1998. He received the M.S. degree in cartography and geographic information systems from Hebei Normal University, Shijiazhuang, China, in 2024.

He is currently working with the Binhai New Area Branch of Tianjin Planning and Natural Resources Bureau. His research interests include geographic information modeling and landscape visual perception computing.